

Bug Fix Impact Analysis

METIMMOX-1 Economic Evaluation

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2026-02-23

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1 Summary

This report compares cost-effectiveness analysis results before and after addressing GitHub issue #143. The comparison assesses whether bug fixes or model changes affected results.

Current Comparison:

- **Issue:** #143 - No title available
- **Baseline:** Commit cd6c906 (2026-02-20 10:58:47)

- **Fixed:** Commit 10533ad (2026-02-20 13:06:44)
- **Models included:** Model A: Joint, Model B: Focused, Model C: Separate

2 Metadata Comparison

Table 1: Metadata Comparison

Item	Baseline	Fixed
Git Commit	cd6c906	10533ad
Timestamp	2026-02-20 10:58:47	2026-02-20 13:06:44
R Version	R version 4.5.1 (2025-06-13 ucrt)	R version 4.3.2 (2023-10-31 ucrt)
Issue Number	143	143
Model Structure	2 (separate)	2 (separate)
Multi-Model Snapshot	Yes	Yes

2.1 Global Parameters

Table 2: Global Parameter Comparison

	Parameter	Baseline	Fixed	Changed
WTP	WTP	51,000	51,000	No
time_horizon	time_horizon	520	520	No
n_samples	n_samples	5,000	5,000	No
n_sim	n_sim	5,000	5,000	No
cl	cl	0.01923077	0.01923077	No
dr	dr	0.04	0.04	No
DSA_mult	DSA_mult	0.2	0.2	No
USE_BOTH_MODELS	USE_BOTH_MODELS	0	0	No
MODEL_STRUCTURE	MODEL_STRUCTURE	2	2	No

3 Per-Model Comparisons

3.1 Model A: Joint

3.1.1 Base Case Results

Table 3: Model A: Joint - Incremental Cost Comparison

Strategy	Baseline (EUR)	Fixed (EUR)	Difference (EUR)
control	0.00	0.00	0.00
crp	34,003.49	34,003.67	0.18
tlr	64,079.67	64,068.27	-11.39
tmb_braf	42,665.89	42,669.90	4.01

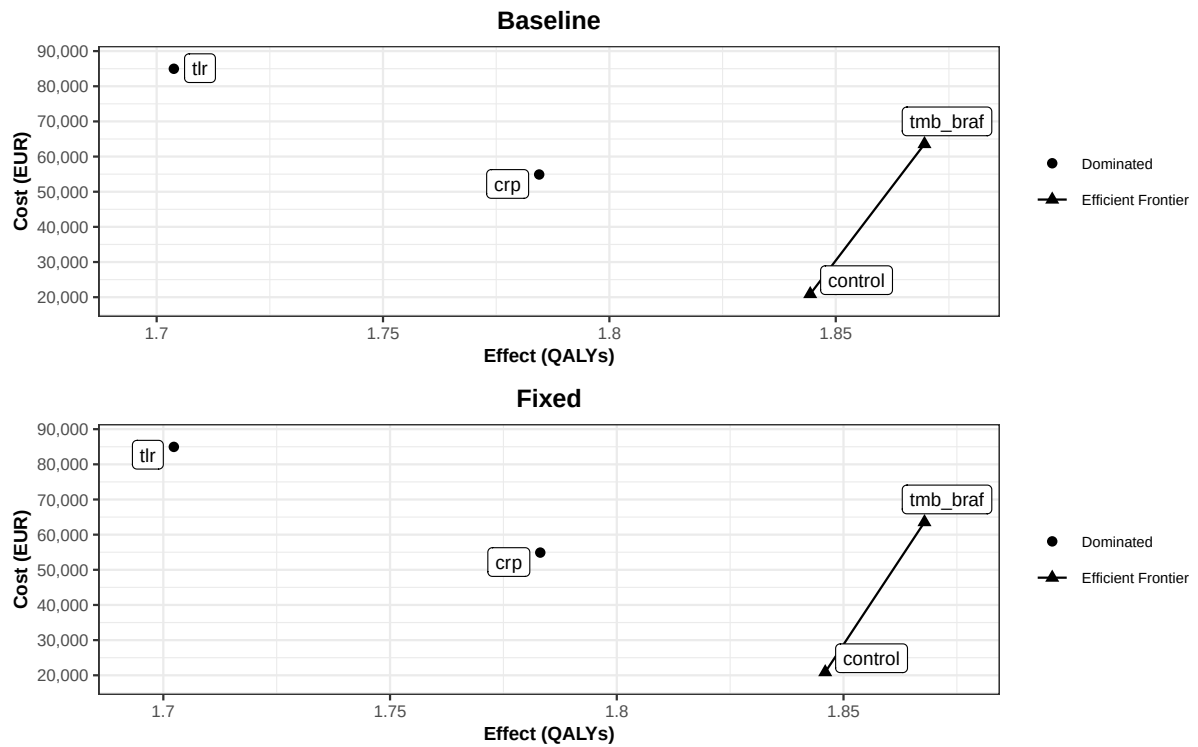
Table 4: Model A: Joint - Incremental Effect Comparison

Strategy	Baseline (QALYs)	Fixed (QALYs)	Difference (QALYs)
control	0.0000	0.0000	0.0000
crp	-0.0599	-0.0628	-0.0030
tlr	-0.1406	-0.1436	-0.0031
tmb_braf	0.0252	0.0219	-0.0033

Table 5: Model A: Joint - ICER Comparison

Strategy	Baseline (EUR/QALY)	Fixed (EUR/QALY)	Difference (EUR/QALY)
control	—	—	—
crp	—	—	—
tlr	—	—	—
tmb_braf	1,690,480	1,949,087	258,607

3.1.2 Efficiency Frontier



TableGrob (2 x 1) “arrange”: 2 grobs z cells name grob 1 1 (1-1,1-1) arrange gtable[layout] 2 2 (2-2,1-1) arrange gtable[layout]

3.1.3 Net Monetary Benefit

Table 6: Model A: Joint - NMB at WTP = EUR 51,000

	Strategy	Baseline (EUR)	Fixed (EUR)	Difference (EUR)
1	control	73,168.05	73,253.51	85.46
2	crp	36,111.38	36,045.09	-66.29
4	tmb_braf	31,789.34	31,700.11	-89.23
3	tlr	1,919.73	1,859.35	-60.38

Optimal Strategy: Unchanged (control)

3.1.4 PSA Results

Table 7: Model A: Joint - PSA Cost Results

Strategy	Baseline Mean (2.5%; 97.5%)	Fixed Mean (2.5%; 97.5%)	Difference (EUR)
control	EUR 21,074.25 (EUR 15,975.03; EUR 27,111.55)	EUR 21,215.88 (EUR 16,169.62; EUR 27,305.36)	141.63
crp	EUR 55,058.21 (EUR 39,626.84; EUR 75,964.95)	EUR 54,947.48 (EUR 39,040.26; EUR 75,202.15)	-110.73
tlr	EUR 85,333.78 (EUR 56,282.73; EUR 122,389.18)	EUR 85,004.32 (EUR 55,713.03; EUR 121,986.42)	-329.46
tmb_braf	EUR 63,490.77 (EUR 45,377.07; EUR 88,712.14)	EUR 63,481.15 (EUR 44,721.07; EUR 87,399.28)	-9.62

Table 8: Model A: Joint - PSA Effect Results

Strategy	Baseline Mean (2.5%; 97.5%)	Fixed Mean (2.5%; 97.5%)	Difference (QALYs)
control	1.6988 (1.0762; 2.3515)	1.7850 (1.1399; 2.5757)	0.0863
crp	1.8097 (1.2080; 2.3102)	1.8136 (1.2488; 2.3133)	0.0039
tlr	1.7537 (1.1176; 2.3700)	1.7545 (1.1394; 2.3548)	0.0007
tmb_braf	1.9296 (1.2485; 2.7611)	1.9349 (1.2716; 2.7840)	0.0054

Table 9: Model A: Joint - PSA ICER Results

Strategy	Baseline Mean ICER	Fixed Mean ICER
control	—	—
crp	EUR 306,223/QALY	EUR 1,178,707/QALY
tlr	EUR 1,168,515/QALY	—
tmb_braf	EUR 183,760/QALY	EUR 281,936/QALY

3.1.5 PSA Scatter Plots



TableGrob (1 x 2) “arrange”: 2 grobs z cells name grob 1 1 (1-1,1-1) arrange gtable[layout] 2 2 (1-1,2-2) arrange gtable[layout]

3.2 Model B: Focused

3.2.1 Base Case Results

Table 10: Model B: Focused - Incremental Cost Comparison

Strategy	Baseline (EUR)	Fixed (EUR)	Difference (EUR)
control	0.00	0.00	0.00
crp	33,857.23	33,857.23	0.00
tlr	65,149.87	65,149.87	0.00
tmb_braf	40,281.45	40,281.45	0.00

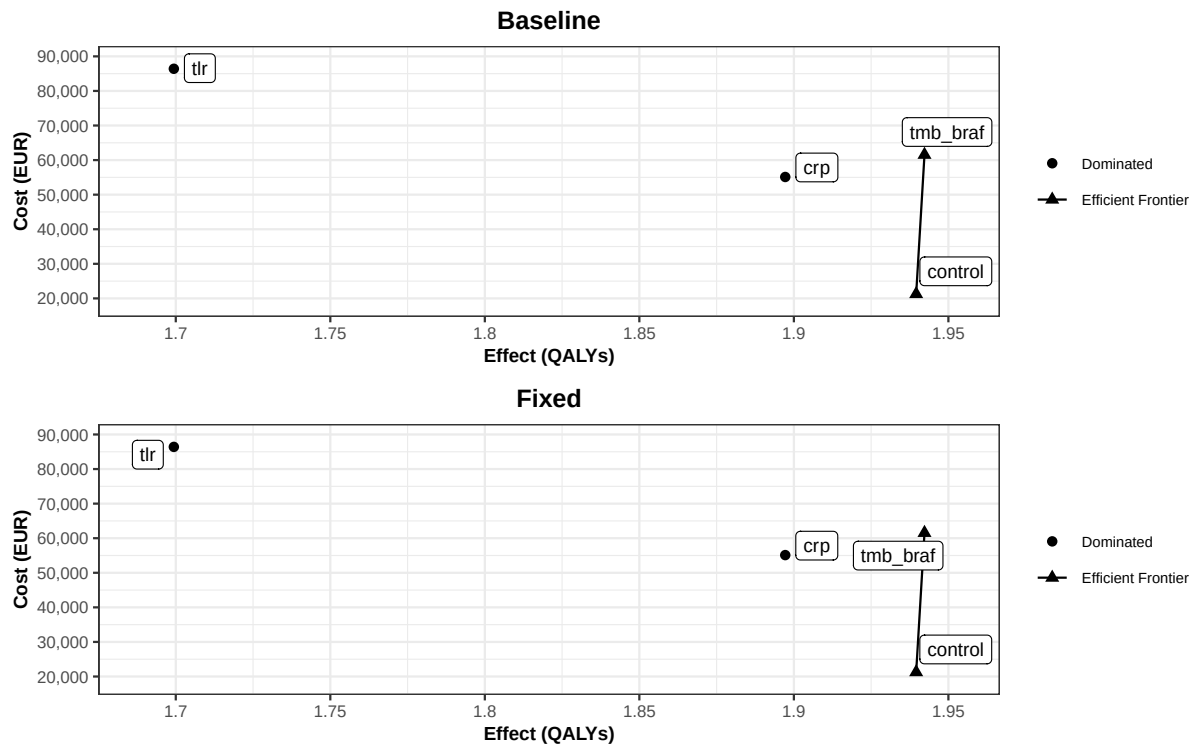
Table 11: Model B: Focused - Incremental Effect Comparison

Strategy	Baseline (QALYs)	Fixed (QALYs)	Difference (QALYs)
control	0.0000	0.0000	0.0000
crp	-0.0424	-0.0424	0.0000
tlr	-0.2402	-0.2402	0.0000
tmb_braf	0.0027	0.0027	0.0000

Table 12: Model B: Focused - ICER Comparison

Strategy	Baseline (EUR/QALY)	Fixed (EUR/QALY)	Difference (EUR/QALY)
control	—	—	—
crp	—	—	—
tlr	—	—	—
tmb_braf	14,980,066	14,980,066	0

3.2.2 Efficiency Frontier



TableGrob (2 x 1) “arrange”: 2 grobs z cells name grob 1 1 (1-1,1-1) arrange gtable[layout] 2 2 (2-2,1-1) arrange gtable[layout]

3.2.3 Net Monetary Benefit

Table 13: Model B: Focused - NMB at WTP = EUR 51,000

	Strategy	Baseline (EUR)	Fixed (EUR)	Difference (EUR)
1	control	77,657.39	77,657.39	0.00
2	crp	41,639.57	41,639.57	0.00
4	tmb_braf	37,513.08	37,513.08	0.00
3	tlr	257.20	257.20	0.00

Optimal Strategy: Unchanged (control)

3.3 Model C: Separate

3.3.1 Base Case Results

Table 14: Model C: Separate - Incremental Cost Comparison

Strategy	Baseline (EUR)	Fixed (EUR)	Difference (EUR)
control	0.00	0.00	0.00
crp	34,292.03	34,292.03	0.00
tlr	66,990.06	66,990.06	0.00
tmb_braf	46,665.00	46,665.00	0.00

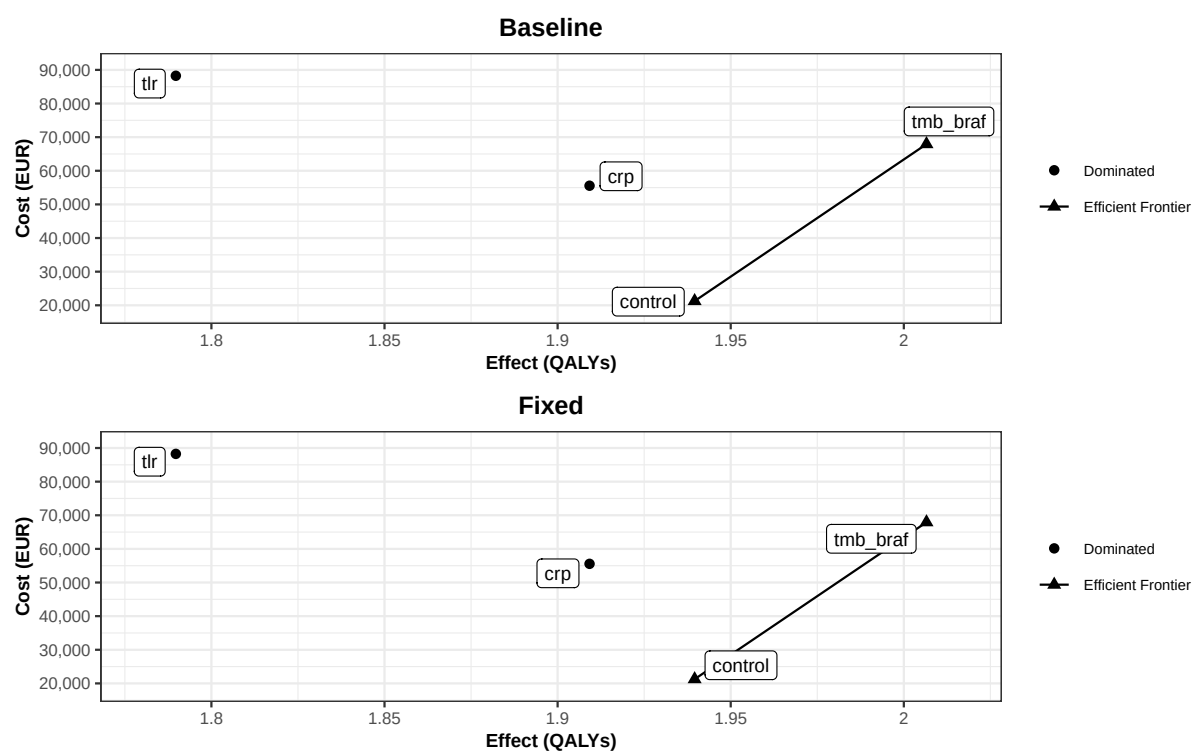
Table 15: Model C: Separate - Incremental Effect Comparison

Strategy	Baseline (QALYs)	Fixed (QALYs)	Difference (QALYs)
control	0.0000	0.0000	0.0000
crp	-0.0304	-0.0304	0.0000
tlr	-0.1498	-0.1498	0.0000
tmb_braf	0.0670	0.0670	0.0000

Table 16: Model C: Separate - ICER Comparison

Strategy	Baseline (EUR/QALY)	Fixed (EUR/QALY)	Difference (EUR/QALY)
control	—	—	—
crp	—	—	—
tlr	—	—	—
tmb_braf	696,842	696,842	0

3.3.2 Efficiency Frontier



TableGrob (2 x 1) “arrange”: 2 grobs z cells name grob 1 1 (1-1,1-1) arrange gtable[layout] 2 2 (2-2,1-1) arrange gtable[layout]

3.3.3 Net Monetary Benefit

Table 17: Model C: Separate - NMB at WTP = EUR 51,000

	Strategy	Baseline (EUR)	Fixed (EUR)	Difference (EUR)
1	control	77,657.39	77,657.39	0.00
2	crp	41,817.15	41,817.15	0.00
4	tmb_braf	34,407.67	34,407.67	0.00
3	tlr	3,025.08	3,025.08	0.00

Optimal Strategy: Unchanged (control)

4 Cross-Model Summary

Table 18: Cross-Model Impact Summary

	Model	Opt. (BL)	Opt. (Fix)	Changed?	Max Cost Delta (EUR)	Max Effect Delta (QALYs)	Max ICER Delta (EUR/QALY)
joint	Model A: Joint	control	control	No	14.94	0.0017	258,607
focused	Model B: Focused	control	control	No	0.00	0.0000	0
separate	Model C: Separate	control	control	No	0.00	0.0000	0

5 Comparison Across Recent Issues

5.1 Available Snapshot Pairs

Table 19: All Available Snapshot Pairs

Issue		Title	Baseline Commit	Fixed Commit
3	143	—	cd6c906	10533ad
2	141	—	9cdec7d	cd6c906
1	118	—	4e55627	10df9fe
4	18	—	71f8230	71f8230
20	99	—	376b4c6	d1eef51
19	97	—	457533e	457533e
7	47	—	2411bdb	2411bdb
6	46	—	e8b3c27	2d5181c
18	94	—	e77415f	3f98095
13	75	—	2b04ea7	c15aa47
17	90	—	9e1b590	7d68bca
14	76	—	64cf543	64cf543
16	87	—	42d4acb	0a15902
15	86	—	ee96c31	f3e5604
12	70	—	998dacd	605ff90
11	68	—	4665f93	998dacd
5	42	—	607b539	cadf81a
10	67	—	f3674e8	d5d1356
8	62	—	d046c7c	f3674e8
9	64	—	8681d1e	8681d1e

5.2 Multi-Issue Impact Summary

This table summarizes the impact of bug fixes across the most recent issues:

Table 20: Summary of Bug Fix Impacts (5 Most Recent Issues)

Issue	Title	Opt. (BL)	Opt. (Fix)	Chg?	Max Cost Delta (EUR)	Max Effect Delta (QALYs)	Max ICER Delta (EUR/QALY)
143	—	control	control	No	14.94	0.0017	258,607
141	—	control	control	No	0.00	0.0000	0
118	—	control	control	No	0.00	0.0000	0
18	—	control	control	No	1,314.18	0.0000	44,593
99	—	control	control	No	0.00	0.0000	0

6 Summary

The comparison for issue #143 shows:

- **Models compared:** Model A: Joint, Model B: Focused, Model C: Separate
- **Optimal Strategy (Model A: Joint):** Unchanged (control)
- **Maximum Cost Change:** EUR 14.94 (minor)
- **Maximum Effect Change:** 0.002 QALYs (minor)
- **Maximum ICER Change:** EUR 258,607/QALY (substantial)
- **Parameter Changes:** None detected

Interpretation: Changes exceed typical thresholds and should be carefully reviewed to ensure they align with the intended bug fix.

Report completed on: 2026-02-23

Repository: ben-geisler/METIMMOX-1

Report version: 3.0